

MOHAMMAD FARAJI

Machine Learning Researcher



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BRIEFLY

I'm a Machine Learning Researcher at the University of Kurdistan working on representation learning, Feature Selection, and Multi-Label learning. Our group is focused on fundamental methodological developments in supervised, semi-supervised, and unsupervised machine learning. Much of our work is related to representation models. I received my Master's in Artificial Intelligence from the Department of Computer, at University of Kurdistan in 2023. My supervisor was Dr. Fardin Akhlaghian and advisor was Amjad Seyedi. My work mostly focused on the matrix factorization and low-rank approximation, predominantly for supervised multi-label feature selection and Multi-label learning problems. Before that, I obtained m Associate and Bachelor's degrees in Software Engineering.

RESEARCH INTERESTS

- Machine learning, Representation learning, Numerical Linear algebra
- Optimization, Low-rank approximation, Matrix Factorization
- Deep learning, Multi-Label learning, Feature Selection

EXPERIENCE

Researcher Full-time	University of Kurdistan, SANANDAJ, IRAN, (Sep 2021 - Present) > skills : Problem solving, Python programming, Data Visualization, and MATLAB.
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EDUCATION

Master	Artificial Intelligence, UNIVERSITY OF KURDISTAN, SANANDAJ, IRAN, (Sep 2020 – Sep 2023) > Thesis title : Multi-Label Feature Selection with Global and Local Label correlation. > Supervisor : Dr. Fardin Akhlaghian. > Courses : machine learning, statistical pattern recognition, neural networks, advanced artificial intelligence, computer vision, digital image processing, complex Networks, and special subjects.
Bachelor	Software Engineering, SHAMSIPOUR TECHNICAL COLLEGE, TEHRAN, IRAN, (Sep 2017 – Jul 2019) > Project title : Student Management system.
Associate	Computer Software, YAZDANPANAHI TECHNICAL COLLEGE, SANANDAJ, IRAN, (Sep 2015 – Sep 2017) > Supplementary courses in computer science and software engineering.
TechSchool	Computer, TALEGHANI HIGH SCHOOL, SANANDAJ, IRAN, (Sep 2013 - Sep 2015) > A two-year education in basic computer science

PUBLICATION

Submitted	Robust log-based multi-label feature selection with dynamic label correlation and relevance-redundancy optimization M. Faraji, S. A. Seyedi, and F. Akhlaghian. Pattern Recognition.
2024	Mutli-Label Feature Selection with Global and Local Label Correlation M. Faraji, S. A. Seyedi, F. Akhlaghian, and R. Mahmoodi. <i>Expert Systems with Applications</i> , volume 246, 2024, pp.110179.

PROFESSIONAL ACTIVITIES

Paper Review

Reviewing and evaluating research manuscripts entitled :
"Carbon Futures Price Forecasting Based on Feature Selection"
Engineering Applications of Artificial Intelligence

Reviewing and evaluating research manuscripts entitled :
"Hybrid Cross-modality Fusion Network for Medical Image Segmentation with Contrastive Learning"
Engineering Applications of Artificial Intelligence

COMPUTER SKILLS

Operating Systems	Microsoft Windows
Word processing & Presentation	Office suites and $\text{\LaTeX}$
Vector and raster softwares	Adobe Illustrator and Adobe XD
Development Tools	Pycharm, Jupyter Notebook, Colab, and Visual Studio
Web design	HTML, CSS, ASP.NET, and JavaScript

PROGRAMMING LANGUAGES

2021 – present	Python, PyTorch, NumPy, Pandas, and scikit-learn
2022 – present	MATLAB, visualization
2015 – 2017	C++ C#, Software Engineering and Web development
2013 – 2015	Basic Visual Basic, Software Engineering

NATURAL LANGUAGES

English	● ● ● ○ ○
Kurdish	● ● ● ● ○
Persian	● ● ● ● ●

+ STRENGTHS

- > Problem-Solving
- > Collaboration and Communication
- > Adaptability and Continuous Learning

REFERENCES

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